

Course name: Mobile Terminal Application Development List and links to videos
Teacher: Dr Ata Jahangir Moshayedi
Project code : <https://github.com/drajm/Android-Studio-20-Projects-to-Build-Your-Skills.git>
Youtube : https://www.youtube.com/watch?v=_MyS93h3nTE&list=PLMNpMrODcRGU



ANDROID STUDIO
 20 Projects to Build Your Skills

#	Project Name	Frontend / UI	Backend / Logic	Key Features	Main Skills / Technologies	Level	Bili Bili link :
1	Hello Android App	XML	Java	Display text, change color/size, show images	Project structure, XML Layout, TextView, ImageView, Button, Activity, Resources	B	【Task 01_ Student Registration Form】 https://www.bilibili.com/video/BV15pdnBWE3j/?share_source=copy_web&vd_source=c2a48a1d55f9b95bff4a352528a4aa93
2	Personal Profile App	XML, ScrollView, LinearLayout	Java, Intents	Profile, photo, email, phone, website buttons	Layouts, Intents, ACTION_SENDTO, ACTION_DIAL, ACTION_VIEW	B	【Task 02_ DASHBOARD】 https://www.bilibili.com/video/BV1EVdnB8EK7/?share_source=copy_web&vd_source=c2a48a1d55f9b95bff4a352528a4aa93
3	Calculator App	XML	Java	Addition, subtraction, multiplication, division	EditText, Button, OnClickListener, Java methods, arithmetic	B	【Task 03_ CONTRACT FORM】 https://www.bilibili.com/video/BV1j7dnBoELD/?share_source=copy_web&vd_source=c2a48a1d55f9b95bff4a352528a4aa93
4	Temperature Converter	XML	Java	°C ↔ °F conversion, validation, formatted results	EditText, inputType, String.format, Toast, conversion formulas	B	【Task 04_ CALCULATOR】 https://www.bilibili.com/video/BV1a8oKBjEgh/?share_source=copy_web&vd_source=c2a48a1d55f9b95bff4a352528a4aa93
5	To-Do List App	XML, RecyclerView	Java	Add, delete, complete tasks; dynamic list	RecyclerView, Adapter, ViewHolder, ArrayList, custom rows	B-	【Task 05_ Personal profile】 https://www.bilibili.com/video/BV1ueoKBCEaJ/?share_source=copy_web&vd_source=c2a48a1d55f9b95bff4a352528a4aa93
6	Notes Application	XML, RecyclerView	Java, Room Database	Create, edit, delete, search, persistent notes	Room, SQLite, Entity, DAO, Repository, ViewModel, LiveData, CRUD	Int	【Task 06_ gallery】 https://www.bilibili.com/video/BV1UeoKBkEYL/?share_source=copy_web&vd_source=c2a48a1d55f9b95bff4a352528a4aa93
7	Quiz Application	XML, Multiple Activities	Java	MCQs, score, timer, progress, multi-screen navigation	Intent, RadioButton, RadioGroup, ProgressBar, CountdownTimer	Int	【Task 07_ navigation bar】 https://www.bilibili.com/video/BV1DYokBVENf/?share_source=copy_web&vd_source=c2a48a1d55f9b95bff4a352528a4aa93
8	Weather Application	XML	Java, Retrofit, REST API	City search, temperature, humidity, wind	REST API, Retrofit, JSON, Gson, Async callbacks, Internet permission	Int	【Task 08_ background color】 https://www.bilibili.com/video/BV19soKBUEHH/?share_source=copy_web&vd_source=c2a48a1d55f9b95bff4a352528a4aa93

9	Expense Tracker App	XML, MPAndroidChart	Java, Room Database	Income/expense, balance, history, monthly charts	Room, RecyclerView, CRUD, financial calculations, PieChart	Int	【Task 09_current date & time】 https://www.bilibili.com/video/BV19soKBUE4c/?share_source=copy_web&vd_source=c2a48a1d55f9b95b5b4a352528a4aa93
10	Camera Application	XML	Java, Camera API	Capture, preview, save images, permissions	Camera API, Runtime Permissions, Intent, Bitmap, MediaStore, Files	Int	【Task10_alert message】 https://www.bilibili.com/video/BV15WoKBnEBu/?share_source=copy_web&vd_source=c2a48a1d55f9b95b5b4a352528a4aa93
11	GPS Location Tracker	XML, Google Maps	Java, Location Manager	Real-time location tracking and map display	Google Maps API, GPS, LocationManager	Int	【Task 11_to do list】 https://www.bilibili.com/video/BV1XtoKB2EGA/?share_source=copy_web&vd_source=c2a48a1d55f9b95b5b4a352528a4aa93
12	QR Code Scanner	XML, CameraX	Java, ML Kit	Scan and decode QR/barcodes	CameraX, ML Kit Barcode Scanning, Camera processing	Int	【Task 12_form validation system】 https://www.bilibili.com/video/BV1XtoKB2EJB/?share_source=copy_web&vd_source=c2a48a1d55f9b95b5b4a352528a4aa93
13	Chat Application	XML, RecyclerView	Firebase	User authentication, real-time messaging	Firebase Auth, Firestore, FCM, RecyclerView	Int	【Task 13_dark mode toggle】 https://www.bilibili.com/video/BV1XtoKB2E4Z/?share_source=copy_web&vd_source=c2a48a1d55f9b95b5b4a352528a4aa93
14	Smart Home Control App	XML	Java, MQTT, ESP32	Control IoT devices via Wi-Fi/Bluetooth	MQTT, ESP32, Bluetooth, Wi-Fi, IoT communication	Adv	【Task 14_dropdown menu】 https://www.bilibili.com/video/BV1V4oKBEEC8/?share_source=copy_web&vd_source=c2a48a1d55f9b95b5b4a352528a4aa93
15	ML Image Classifier	XML, Camera	Java, TensorFlow Lite	Real-time image classification	CNN, TensorFlow Lite, Camera API, GPU acceleration	Adv	【Task 15_Tab-Based Content System】 https://www.bilibili.com/video/BV1V4oKBEEWA/?share_source=copy_web&vd_source=c2a48a1d55f9b95b5b4a352528a4aa93
16	Face Recognition App	XML, Camera	Java, OpenCV/CNN	Face detection and recognition	OpenCV, Face Embeddings, CNN, Camera	Adv	【Task 16_Quiz system】 https://www.bilibili.com/video/BV1bRoNBgEHK/?share_source=copy_web&vd_source=c2a48a1d55f9b95b5b4a352528a4aa93
17	Voice Assistant App	XML	Java, Speech/TTS	Voice commands and responses	Speech Recognition, Text-to-Speech, Intent System	Int-Adv	【Task 17_image slider】 https://www.bilibili.com/video/BV1uRoNBgEAp/?share_source=copy_web&vd_source=c2a48a1d55f9b95b5b4a352528a4aa93
18	Medical Image Analysis	XML, Camera/Gallery	Java, TensorFlow Lite	Medical image segmentation/analysis	U-Net, TensorFlow Lite, GPU Delegate, Image Processing	Adv	【Task 18_Model popup】 https://www.bilibili.com/video/BV1WNoKBYELw/?share_source=copy_web&vd_source=c2a48a1d55f9b95b5b4a352528a4aa93
19	Autonomous Robot Control	XML, Video Interface	Java, ESP32	Robot control, video streaming, remote navigation	Wi-Fi/Bluetooth, ESP32, Video Streaming, Robotics	Adv	【Task 19_search filter】 https://www.bilibili.com/video/BV1ANoKBYECs/?share_source=copy_web&vd_source=c2a48a1d55f9b95b5b4a352528a4aa93



20	AI + IoT Smart Healthcare	Kotlin, Jetpack Compose	Firebase, REST, MQTT, AI/IoT	Full-stack intelligent healthcare system	Kotlin, Compose, Firebase, REST, MQTT, AI, IoT	Adv	【Task 20 _ progress bar】 https://www.bilibili.com/video/BV1ANoKBYE2o/?share_source=copy_web&vd_source=c2a48a1d55f9b95bfff4a352528a4aa93
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